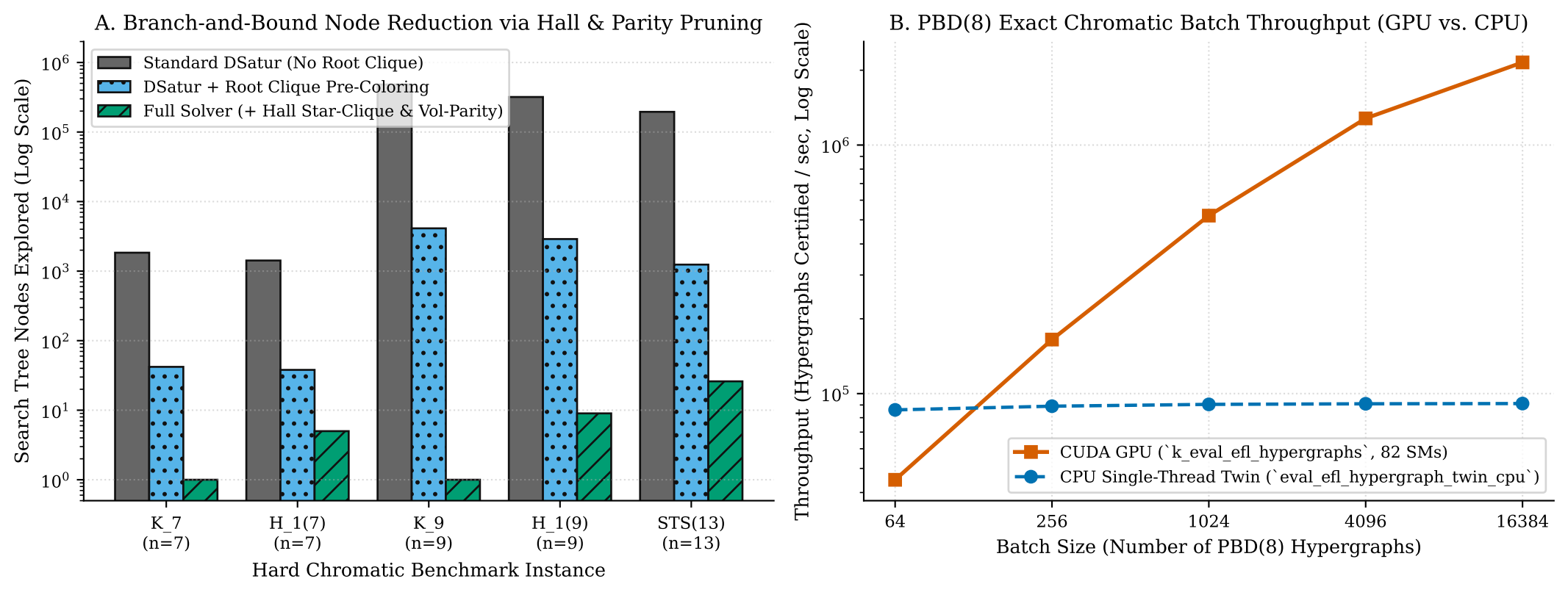
A. Branch-and-Bound Node Reduction via Hall & Parity Pruning
Standard DSatur (No Root Clique)
DSatur + Root Clique Pre-Coloring
Full Solver (+ Hall Star-Clique & Vol-Parity)
Search Tree Nodes Explored (Log Scale)
10^0
10^1
10^2
10^3
10^4
10^5
10^6
K_7 (n=7)
H_1(7) (n=7)
K_9 (n=9)
H_1(9) (n=9)
STS(13) (n=13)
Hard Chromatic Benchmark Instance
B. PBD(8) Exact Chromatic Batch Throughput (GPU vs. CPU)
Throughput (Hypergraphs Certified / sec, Log Scale)
10^5
10^6
CUDA GPU (`k_eval_efl_hypergraphs`, 82 SMs)
CPU Single-Thread Twin (`eval_efl_hypergraph_twin_cpu`)
64
256
1024
4096
16384
Batch Size (Number of PBD(8) Hypergraphs)